

Loading the battery dataset

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%% STEP 1: Download and Load Battery Dataset
dataFile
= matlab.internal.examples.downloadSupportFile("predmaint","batteryagingdata/
singlecell/v1/singleCellLifeTimeData.zip");
unzip(dataFile);
load("singleCellLifeTimeData.mat");

%% STEP 2: Parse and Segment Data
bParser = batteryTestDataParser(data);
bParser.CurrentVariable = 'Current';
bParser.VoltageVariable = 'Voltage';
bParser.TimeVariable = 'DateTime';
bParser.CycleIndexVariable = 'Cycle_Index';
bParser.StepIndexVariable = 'Step_Index';
segmentedRawDataTable = segmentData(bParser);

%% STEP 3: Extract All Valid "Charge" Data for Cycle 1
chargingData = segmentedRawDataTable(...
    segmentedRawDataTable.Cycle_Index == 1 & ...
    segmentedRawDataTable.CyclingPhases == "Charge", :);
chargingData.TimeInSeconds = seconds(chargingData.DateTime -
chargingData.DateTime(1));
```

The total energy delivered to the battery during charging

Use integration to calculate the area under the power-time curve.

```
% 1. Pull time t and power P from charging data
t = chargingData.TimeInSeconds;
P = chargingData.Current .* chargingData.Voltage;

% 2. Trapezoidal integration to compute energy in Joules (Watt-Seconds)
total_energy_joules = trapz(t, P);

% 3. Convert Joules to Watt-hours (1 Wh = 3600 Joules) for standard battery metrics
total_energy_wh = total_energy_joules / 3600;

% 4. Display Results
fprintf('Total Energy Delivered: %.2f Joules (J)\n', total_energy_joules);
```

Total Energy Delivered: 14221.61 Joules (J)

```
fprintf('Total Energy Delivered: %.4f Watt-hours (Wh)\n', total_energy_wh);
```

Total Energy Delivered: 3.9504 Watt-hours (Wh)